

Assignment: Programming Strategies

Type: Group Assignment

Important Rules and Penalty Policy

- Students should **ONLY USE** programming constructs covered in units 1 and 2.
- **Using programming concepts that are not covered in the above units will be penalized.**
 - **Penalty with no limit could be applied**
 - **Minimum penalty is 25%**
- **Late submissions will not be accepted**

Scenario

AZ Financial Institution hired a software company to write a program to calculate monthly mortgage payments and produce amortization schedule for a selected mortgage terms. The program uses authentic rules and rates applicable to current British Columbia real estate purchases, which are described below in the Mortgage Calculator Details section.

Equipment and Materials

For this assignment, you will need:

- Python IDE

Instructions

This assignment will need to be completed outside of class time. See the course schedule and Brightspace for exact due dates. There are many details involved in this assignment which could be challenging to get sorted out and working correctly. Therefore, it is strongly recommended to approach the problem in two stages, by focusing on and completing Part 1 before attempting Part 2.

Group Submission

1. Working in a group, review the Scenario and Mortgage Calculator Details sections of this document, noting how the program functionality changes between Part 1 and Part 2.
2. Review the grading criteria for submission.
3. Draft a pseudo code or flowchart to represent **Part 1** of the program (**No Submission**).
4. Write the code for a program that meets the requirements outlined for **Part 1**.

5. Test your code using the same input values as shown in the **Part 1** Monthly Payment Calculator working example to verify that your results are correct.
6. Modify your Part 1 pseudo code or flowchart to include **Part 2 (No Submission)**.
7. Make a copy of your Part 1 code and extend it to meet the **Part 2** requirements.
8. Test your code against the expected output provided for **Part 2**.
9. Submit the following to Brightspace:
 - The final code of the program that you implemented (.py file) for **Part 2**. You should submit only Part 2 solution unless you were unable to complete Part 2, in which case submit your completed Part 1 solution.
 - Your test outputs for **Part 2** (.txt files), covering all 5 test cases.
 - The PDF/DOC peer assessment according to your instructor's instructions.

AZ Financial Institution Mortgage Calculator Details

The program can be gradually developed using 2 parts as follows:

Part 1: Monthly Payment Calculator (No submission because Part 1 will be extended to implement Part 2)

1. Minimum Down Payment Component:

The initial user inputs to the calculator are the client's name, address, and purchase price of the property to be mortgaged. The next input is the down payment percentage; however, this value must meet or exceed a minimum that varies depending on the purchase price as shown in table 1 below.

Table 1: Minimum Down Payment Percentage

Purchase Price	Minimum Down Payment
<= \$400,000	5%
> \$400,000 and <= \$900,000	5% on \$400,000 + 10% on the remainder
> \$900,000	20%

Example: For a property with a purchase price of \$800,000 the minimum down payment amount would be calculated as follows:

Recalculate if you changed the above values

5% on \$400,000 + 10% on the remainder of \$400,000

\$20,000 + \$40,000 = \$60,000

Expressing this minimum down payment as a percentage of the purchase price gives us:

$\$60,000 / \$800,000 * 100 = 7.5\%$

2. Mortgage Default Insurance Component

Once the down payment percentage user input is received, the calculator determines the applicable mortgage default (CMHC) insurance rate using the rules shown in table 2 below. With these values determined it is now simple to calculate down payment amount, mortgage insurance cost, and then the principal amount.

Table 2: Required Mortgage Default Insurance

Down Payment	Mortgage Insurance Premium (rate applied to loan amount)
5% up to but not including 10%	5%
10% up to but not including 15%	4%
15% up to but not including 20%	3%
20% and up	0% (i.e. no insurance required)

Continuing with the previous example and assuming the desired down payment is 10%, the down payment amount, mortgage insurance cost, and resulting principal (total mortgage) amount are determined as follows:

Recalculate

$$\begin{aligned}\text{Down payment amount} &= \text{purchase price} * \text{down payment percentage} / 100 \\ &= \$800,000 * 10 / 100 = \$80,000\end{aligned}$$

$$\begin{aligned}\text{Insurance cost} &= (\text{purchase price} - \text{down payment amount}) * \text{insurance rate} / 100 \\ &= (\$800,000 - \$80,000) * 4 / 100 = \$28,800\end{aligned}$$

$$\begin{aligned}\text{Principal amount} &= \text{purchase price} - \text{down payment amount} + \text{insurance cost} \\ &= \$800,000 - \$80,000 + \$28,800 = \$748,800\end{aligned}$$

3. Mortgage Payment Component

The final two user inputs to part 1 of the mortgage calculator are the mortgage term and the mortgage amortization period. The following tables list the allowable values for both of these inputs. Each mortgage term value is accompanied by an applicable mortgage interest rate charged by AZ Financial Institution.

Table 3: Mortgage Terms and Applicable Mortgage Interest Rates

Mortgage Term (years)	Mortgage Interest Rate
1	5.9%
2	5.8%

3	5.5%
5	5.3%
10	6%

Table 4: Mortgage Amortization Periods

Amortization Period (years)
5
10
15
20
25

To conclude part 1 of the calculator, the effective monthly interest rate and monthly payment amount are calculated using the formulas shown below:

- $EMR = ((1 + A/2)^2)^{1/12} - 1$

Where:

A is the Annual mortgage interest rate (as per table 3 above)

EMR is the Effective Monthly Rate

- $M = P[EMR(1 + EMR)^n] / [(1 + EMR)^n - 1]$

Where:

M is the Monthly payment

P is the Principal amount

EMR is the Effective Monthly Rate as previously calculated

n is the number of monthly payments (years * 12)

Continuing with our same example, let's assume the input values for mortgage term and amortization period are 10 years and 15 years respectively. This gives us:

$$A = 0.06 \quad (\text{i.e. 6\% rate for the chosen 10-year term})$$

$$EMR = ((1 + 0.06/2)^2)^{1/12} - 1$$

$$EMR = ((1 + 0.03)^2)^{1/12} - 1$$

$$EMR = (1.0609)^{1/12} - 1$$

$$EMR = 1.00493862 - 1$$

$$\text{EMR} = 0.00493862$$

For a \$748,800 mortgage amount at 6% compounded semi-annually with an amortization period of 15 years:

$$n = 12 \times 15 = 180 \text{ monthly payments}$$

$$M = 748800[0.00493862(1 + 0.00493862)^{180}] / [(1 + 0.00493862)^{180} - 1]$$

$$M = 748800[(0.00493862 * 2.42726159) / (2.42726159 - 1)]$$

$$M = 748800[0.01198732] / 1.42726159$$

$$M = \$6,289.04$$

Important: After successfully completing Part 1, save a copy of your work for your reference.

Part 2: The Amortization Schedule

In addition to calculating monthly payments, AZ Financial Institution would like the program to optionally print out a monthly amortization schedule. This schedule, or report, shows the opening balance, monthly payment, monthly interest (i.e. portion of payment going to interest), monthly principal (i.e. portion of payment going to principal), and closing balance, for each month of the mortgage term.

Extend your program from Part 1 so that it can calculate and print an amortization schedule using the logic outlined below.

To figure out how much of the monthly payment will be applied to the principle and how much needs to be paid to AZ Financial Institution as interest we use the following formulas:

- **MI = OB * EMR**
- **MP = M – MI**
- **CB = OB – MP**

Where:

OB is Opening Balance at the start of the month

EMR is Effective Monthly Rate (as previous calculated)

M is Monthly payment (as previously calculated)

MP is Monthly Principle amount

MI is Monthly Interest amount

CB is Closing Balance at the end of the month

Plugging our example values into these formulas produces the following results for the first month:

$$\text{OB} = 748,800$$

$$\text{EMR} = 0.00493862$$

$$M = 6,289.04$$

$$MI = OB * EMR$$

$$MI = 748,800 * 0.00493862$$

$$MI = 3,698.04 \text{ (Monthly Interest)}$$

$$MP = M - MI$$

$$MP = 6289.04 - 3,698.04$$

$$MP = 2,591$$

$$CB = OB - MP$$

$$CB = 748,800 - 2,591$$

$$CB = 746,209$$

Other Requirements:

- The program needs to validate the following **three** user inputs:
 - Down payment must be between the minimum (from table 1) and 100
 - Mortgage term must be one of the values from table 3
 - Amortization period must be one of the values from table 4

If a value entered by the user falls outside of the allowed values, display an appropriate message, and allow them to try again. The program should not move on to the next step until a valid value has been input.

- It is NOT necessary to verify that input values match the expected data type. In other words, you can assume that the user will enter digits for an integer item such as mortgage term.
- When calculating minimum down payment percentages that have a combination of the 5% and 10% components, you should round the resulting percentage to 3 digits after the decimal place.

Tips:

- Remember to use good variable names. The initialisms used as variables in the preceding formulas are convenient for this document, but they are NOT good Python variable names.
- Use f-strings to format output. Research how to use f-strings to align columns and restrict the number of digits output after the decimal point.
- Use constants for predetermined values such as percentages/rates and available terms.
- Consider how you might be able to use compound data types to streamline your solution.
- Use one of the following online calculators and compare the results with your program's output:

- <https://www.cmhc-schl.gc.ca/en/consumers/home-buying/calculators/mortgage-calculator>
- <https://www.realtor.ca/calculator>

Teamwork Strategy

- Assign a team member to implement and test each task.
- Once development and testing of a task is complete, other members should review the code.
- After completing all tasks, assign a team member to integrate all the code together into a single solution.
- Assign a team member to add documentation.
- Assign a team member to review the whole code.
- The group members test the whole program and submit source code and sample runs to Brightspace.

Part 2 Test Plan

Run your program for the 5 sample runs provided. Input values are shown in **bold underline**.

Sample Run # 1

Enter client name: **Maria Smith**
Enter address of property: **20 ABC Close**
Enter purchase price: **315000**
Enter down payment percentage (minimum 5): **10**
Down payment amount is \$31500
Mortgage insurance price is \$11340
Total mortgage amount is \$294840
Enter mortgage term (1, 2, 3, 5, 10): **1**
Enter mortgage amortization period (5, 10, 15, 20, 25): **20**
Interest rate for the term will be 5.90%
Monthly payment amount is: \$2083.30
Would you like to see the amortization schedule? (Y/N): **Y**
Monthly Amortization Schedule

Month	Opening Bal	Payment	Principal	Interest	Closing Bal
1	294840.00	2083.30	651.18	1432.13	294188.82
2	294188.82	2083.30	654.34	1428.96	293534.49
3	293534.49	2083.30	657.52	1425.79	292876.97
4	292876.97	2083.30	660.71	1422.59	292216.26

5	292216.26	2083.30	663.92	1419.38	291552.34
6	291552.34	2083.30	667.14	1416.16	290885.19
7	290885.19	2083.30	670.39	1412.92	290214.81
8	290214.81	2083.30	673.64	1409.66	289541.17
9	289541.17	2083.30	676.91	1406.39	288864.25
10	288864.25	2083.30	680.20	1403.10	288184.05
11	288184.05	2083.30	683.51	1399.80	287500.54
12	287500.54	2083.30	686.83	1396.48	286813.72
=====					
Total			8026.28	16973.34	

Sample Run # 2

Enter client name: John Mike
Enter address of property: 2000 XYZ DR
Enter purchase price: 500500
Enter down payment percentage (minimum 6.004): 5
Please enter a value between the minimum and 100
Enter down payment percentage (minimum 6.004): 101.5
Please enter a value between the minimum and 100
Enter down payment percentage (minimum 6.004): 6.004
Down payment amount is \$30050
Mortgage insurance price is \$23522
Total mortgage amount is \$493972
Enter mortgage term (1, 2, 3, 5, 10): 10
Enter mortgage amortization period (5, 10, 15, 20, 25): 25
Interest rate for the term will be 6.00%
Monthly payment amount is: \$3160.47
Would you like to see the amortization schedule? (Y/N): n

Sample Run # 3

Enter client name: Lisa Smith
Enter address of property: 90 KLM St.
Enter purchase price: 670000
Enter down payment percentage (minimum 7.015): 30
Down payment amount is \$201000
Mortgage insurance price is \$0
Total mortgage amount is \$469000
Enter mortgage term (1, 2, 3, 5, 10): 5
Enter mortgage amortization period (5, 10, 15, 20, 25): 5
Interest rate for the term will be 5.30%
Monthly payment amount is: \$8902.79
Would you like to see the amortization schedule? (Y/N): Y

Monthly Amortization Schedule

Month	Opening Bal	Payment	Principal	Interest	Closing Bal
1	469000.00	8902.79	6853.88	2048.91	462146.12
2	462146.12	8902.79	6883.82	2018.97	455262.30
3	455262.30	8902.79	6913.90	1988.89	448348.40
4	448348.40	8902.79	6944.10	1958.69	441404.30
5	441404.30	8902.79	6974.44	1928.35	434429.86
6	434429.86	8902.79	7004.91	1897.88	427424.95
7	427424.95	8902.79	7035.51	1867.28	420389.44
8	420389.44	8902.79	7066.24	1836.54	413323.20

9	413323.20	8902.79	7097.11	1805.67	406226.08
10	406226.08	8902.79	7128.12	1774.67	399097.96
11	399097.96	8902.79	7159.26	1743.53	391938.70
12	391938.70	8902.79	7190.54	1712.25	384748.17
13	384748.17	8902.79	7221.95	1680.84	377526.22
14	377526.22	8902.79	7253.50	1649.29	370272.72
15	370272.72	8902.79	7285.19	1617.60	362987.53
16	362987.53	8902.79	7317.02	1585.77	355670.51
17	355670.51	8902.79	7348.98	1553.81	348321.53
18	348321.53	8902.79	7381.09	1521.70	340940.45
19	340940.45	8902.79	7413.33	1489.46	333527.12
20	333527.12	8902.79	7445.72	1457.07	326081.40
21	326081.40	8902.79	7478.25	1424.54	318603.15
22	318603.15	8902.79	7510.92	1391.87	311092.24
23	311092.24	8902.79	7543.73	1359.06	303548.51
24	303548.51	8902.79	7576.68	1326.10	295971.82
25	295971.82	8902.79	7609.78	1293.00	288362.04
26	288362.04	8902.79	7643.03	1259.76	280719.01
27	280719.01	8902.79	7676.42	1226.37	273042.59
28	273042.59	8902.79	7709.96	1192.83	265332.63
29	265332.63	8902.79	7743.64	1159.15	257589.00
30	257589.00	8902.79	7777.47	1125.32	249811.53
31	249811.53	8902.79	7811.44	1091.35	242000.08
32	242000.08	8902.79	7845.57	1057.22	234154.51
33	234154.51	8902.79	7879.84	1022.94	226274.67
34	226274.67	8902.79	7914.27	988.52	218360.40
35	218360.40	8902.79	7948.84	953.95	210411.56
36	210411.56	8902.79	7983.57	919.22	202427.99
37	202427.99	8902.79	8018.45	884.34	194409.54
38	194409.54	8902.79	8053.48	849.31	186356.06
39	186356.06	8902.79	8088.66	814.13	178267.40
40	178267.40	8902.79	8124.00	778.79	170143.40
41	170143.40	8902.79	8159.49	743.30	161983.92
42	161983.92	8902.79	8195.13	707.66	153788.78
43	153788.78	8902.79	8230.94	671.85	145557.85
44	145557.85	8902.79	8266.89	635.89	137290.95
45	137290.95	8902.79	8303.01	599.78	128987.94
46	128987.94	8902.79	8339.28	563.51	120648.66
47	120648.66	8902.79	8375.71	527.07	112272.94
48	112272.94	8902.79	8412.31	490.48	103860.64
49	103860.64	8902.79	8449.06	453.73	95411.58
50	95411.58	8902.79	8485.97	416.82	86925.61
51	86925.61	8902.79	8523.04	379.75	78402.57
52	78402.57	8902.79	8560.27	342.52	69842.30
53	69842.30	8902.79	8597.67	305.12	61244.63
54	61244.63	8902.79	8635.23	267.56	52609.40
55	52609.40	8902.79	8672.96	229.83	43936.44
56	43936.44	8902.79	8710.85	191.94	35225.60
57	35225.60	8902.79	8748.90	153.89	26476.70
58	26476.70	8902.79	8787.12	115.67	17689.57
59	17689.57	8902.79	8825.51	77.28	8864.07
60	8864.07	8902.79	8864.07	38.72	-0.00
=====					
Total			469000.00	65167.37	

Sample Run # 4

Enter client name: Angle Miracle
 Enter address of property: 3000 Rocky Mountain BLVD
 Enter purchase price: 999999
 Enter down payment percentage (minimum 20): 25
 Down payment amount is \$250000
 Mortgage insurance price is \$0
 Total mortgage amount is \$749999
 Enter mortgage term (1, 2, 3, 5, 10): 3
 Enter mortgage amortization period (5, 10, 15, 20, 25): 17
 Please enter a valid choice
 Enter mortgage amortization period (5, 10, 15, 20, 25): 15
 Interest rate for the term will be 5.50%
 Monthly payment amount is: \$6103.48
 Would you like to see the amortization schedule? (Y/N): N

Sample Run # 5

Enter client name: H. I. Madien
 Enter address of property: 170 International Road
 Enter purchase price: 1420000
 Enter down payment percentage (minimum 20): 20
 Down payment amount is \$284000
 Mortgage insurance price is \$0
 Total mortgage amount is \$1136000
 Enter mortgage term (1, 2, 3, 5, 10): 22
 Please enter a valid choice
 Enter mortgage term (1, 2, 3, 5, 10): 2
 Enter mortgage amortization period (5, 10, 15, 20, 25): 10
 Interest rate for the term will be 5.80%
 Monthly payment amount is: \$12459.09
 Would you like to see the amortization schedule? (Y/N): Y

Monthly Amortization Schedule

Month	Opening Bal	Payment	Principal	Interest	Closing Bal
1	1136000.00	12459.09	7033.62	5425.47	1128966.38
2	1128966.38	12459.09	7067.21	5391.88	1121899.16
3	1121899.16	12459.09	7100.97	5358.13	1114798.20
4	1114798.20	12459.09	7134.88	5324.21	1107663.32
5	1107663.32	12459.09	7168.96	5290.14	1100494.36
6	1100494.36	12459.09	7203.19	5255.90	1093291.17
7	1093291.17	12459.09	7237.60	5221.50	1086053.57
8	1086053.57	12459.09	7272.16	5186.93	1078781.41
9	1078781.41	12459.09	7306.89	5152.20	1071474.51
10	1071474.51	12459.09	7341.79	5117.30	1064132.72
11	1064132.72	12459.09	7376.86	5082.24	1056755.87
12	1056755.87	12459.09	7412.09	5047.01	1049343.78
13	1049343.78	12459.09	7447.49	5011.61	1041896.29
14	1041896.29	12459.09	7483.06	4976.04	1034413.24
15	1034413.24	12459.09	7518.79	4940.30	1026894.44
16	1026894.44	12459.09	7554.70	4904.39	1019339.74
17	1019339.74	12459.09	7590.78	4868.31	1011748.95
18	1011748.95	12459.09	7627.04	4832.06	1004121.92
19	1004121.92	12459.09	7663.46	4795.63	996458.45

20	996458.45	12459.09	7700.06	4759.03	988758.39
21	988758.39	12459.09	7736.84	4722.26	981021.55
22	981021.55	12459.09	7773.79	4685.30	973247.76
23	973247.76	12459.09	7810.92	4648.18	965436.84
24	965436.84	12459.09	7848.22	4610.87	957588.62
=====					
Total			178411.38	120606.90	

Peer Assessment

- Only one DOC/PDF should be used for peer assessment.
- Final grades will be assigned according to members' participation.
- The peer assessment should be completed accurately and honestly
 - Deduction will be applied if it is not
- Each member should assess the contribution/participation of all other group members.
- Each member should assign a mark out of 10 to other members.
- Accurately describe the task completed by each member
 - Deduction will be applied if it is not
- Please use the following table to complete the peer assessment. Marks/tasks are just samples
 - Deduction will be applied if it is not used

	Member 1 Name	Member 2 Name	Member 3 Name	Completed Tasks
Member 1 Name	N/A	9	10	• Task 1 description • Task 2 description
Member 2 Name	8	N/A	9	• Task 3 description • Task 4 description
Member 3 Name	9	10	N/A	• Task 5 description • Task 6 description
Average	8.5	9.5	9.5	•

Note that:

- A student can receive a deduction on their individual grade for the assignment if they do not meet the expected contribution/participation based on the feedback of other group members.

Marking Criteria

Rubric available on Brightspace.